

Static Testing Report

Static testing tool(s) reports/screenshots and brief explanation of results.

Tool

SonarQube/SonarCloud was used as the static testing tool. It analyzed the code without running it and reported maintainability issues as code smells.

Screenshots

Use `assertThat(actual).contains(expected)` instead.

Intentionality

Maintainability Low

assertj tests

☐ Open Ahmad Alkaseb

L29 · 5min effort · 14 days ago · Code Smell · Minor

Use `"isEmpty()"` to check whether a `"String"` is empty or not.

Intentionality

Maintainability Low

No tags

☐ Open Ahmad Alkaseb

L37 · 2min effort · 14 days ago · Code Smell · Minor

```
118 - TypedQuery<T> query = em.createQuery("SELECT a FROM " + entityClass.getSimpleName() + " a WHERE a.id = :id",
119 - entityClass);
120 - query.setParameter("id", id);
121 - return query.getSingleResult();
122 - } catch (NoResultException e) {
123 -     throw new EntityNotFoundException(timestamp + ": No " + entityClass.getSimpleName() + " found with id: " + id);
124 - }
125 + T entity = em.find(entityClass, id);
126 + if (entity == null) {
127 +     throw new EntityNotFoundException(timestamp + ": No " + entityClass.getSimpleName() + " found with id: " + id);
128 + }
129 + return entity;
```

Results

- The reported issues were Minor Code Smells, not runtime failures.
- Two findings were low-effort maintainability improvements in tests: use AssertJ contains directly and use isEmpty() for empty string checks.
- The DAO finding led to simpler persistence code: the manual TypedQuery/getSingleResult logic was replaced with EntityManager.find(...).
- Overall result: the static analysis helped clean up readability, reduce unnecessary code, and make the test assertions easier to understand.

Brief explanation

The static report was useful because it pointed out small but concrete maintainability problems. The issues did not show broken functionality, but they improved code quality by making assertions more idiomatic and replacing custom query code with a simpler JPA method.